

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building AI-driven EDA systems that automate synthesis, verification, and physical design. Built SiliconCrew — an ML-powered agent that drives RTL through synthesis, placement, routing, and signoff, scoring **37/92 on CVDP** and **5/5 on ASU Spec2Tapeout** with zero human intervention, with a full MCP server (stdio/SSE/HTTP) for tool integration. Designed a GNN ASIC accelerator end-to-end on ASAP7 7nm (Synopsys DC + Cadence Innovus).

TECHNICAL SKILLS

AI / ML for EDA: Python | PyTorch | TensorFlow | LangGraph | MCP Protocol | Multi-provider LLM Routing | RAG Pipelines | Prompt Engineering | CUDA

EDA Tools & Flows: Synopsys Design Compiler | Cadence Innovus | Voltus | OpenROAD | Yosys | SymbiYosys | Static Timing Analysis | NLDM Liberty | Tcl (SDC)

RTL & Architecture: SystemVerilog | Verilog | Cocotb | iverilog | C++ | ASAP7 7nm | SkyWater 130nm | gem5 | Docker | Linux

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

May 2026

Tempe, AZ

Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — AI-Driven EDA Automation with MCP Integration

Nov 2025 – Present

- Built SiliconCrew, an ML-powered agent (21 tools) that automates the end-to-end EDA flow — RTL generation → lint → simulation → Yosys synthesis → OpenROAD placement and routing → post-synthesis simulation → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs; built a full MCP server (stdio/SSE/HTTP) for Claude Desktop and VS Code; curates simulation outputs, VCD waveforms, and synthesis reports into structured datasets for iterative agent improvement.
- Closes the gap between AI-driven EDA and sign-off: on simulation failure, reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies autonomously; enforces SDC constraint validation, RTL-vs-netlist equivalence checking, and signoff guardrails — every simulation cycle yields actionable diagnostic data for the next iteration.
- Achieved **37/92 first-pass on CVDP** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout** with zero human intervention; multi-provider ML inference routing across Gemini, OpenAI, and Anthropic with provider-specific optimization.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm

Spring 2025

- Designed an 11-module Graph Neural Network ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved 50,300 μm^2 chip area at 50.6% core density, 52.29 mW total power (64.8% switching), and 99.2 ns end-to-end latency.

CUDA MLP — Neural Network Training on GPU

Fall 2024

- Implemented 6 CUDA kernels in C++ (linear forward/backward, ReLU forward/backward) for MLP training on MNIST — compiled via nvcc to shared libraries, bound to Python/PyTorch through ctypes FFI, with manual cudaMalloc/cudaMemcpy buffer management; end-to-end ML training pipeline from raw data to converged model.

Static Timing Analysis Engine — Python

Fall 2025 – Spring 2026

- Built a complete Static Timing Analysis engine in Python from scratch — Liberty NLDM parser (7×7 delay/slew LUTs with bilinear interpolation), .bench circuit-graph parser, topological sort via Kahn's algorithm, forward arrival-time and slew propagation, backward RAT/slack computation, and critical path extraction; validated against ISCAS benchmarks c17, b15, and c7552.

Trezzit — Production ML System (trezzit.com)

Feb 2025 – Present

- Sole-built and shipped a production ML system serving **600+ users** for 12+ months — multi-provider inference pipeline (Gemini, OpenAI, Claude, Grok) that improved parsing accuracy from ~60% → **95%+** via two-pass semantic validation with automatic model-upgrade fallback; **995 tests**, GitHub Actions CI/CD, data pipeline tracking per-provider cost and latency in production.

WORK EXPERIENCE

Systems Engineer

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

Research Intern — Propulsion & Power Electronics

Oct 2021 – Aug 2022

Indian Institute of Science

Bengaluru, India

- Applied Ansys Maxwell FEA to optimize a 1.5 kW axial flux BLDC motor — redesigned stator geometry and upgraded magnetic materials to improve flux linkage and reduce core losses; modeled FOC control and battery management in Simulink for eVTOL propulsion reliability.

AWARDS & HONORS

- Won **‘Best Use of AI’** at Strategy X DevHacks’25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.